TANVIR HASAN RATUL

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LinkedIn | GitHub | LeetCode

OBJECTIVE

Computer Science undergraduate seeking a Software Engineering or Backend Developer internship. Experienced building FastAPI backends, relational systems, REST APIs, and offline-capable React applications; comfortable with SQL, object-oriented programming, authentication, testing, and data tools.

EDUCATION

Ahsanullah University of Science and Technology (AUST) — 3rd Year, 2nd Semester
B.Sc. in Computer Science and Engineering — **CGPA: 3.223 / 4.00**

St. Joseph Higher Secondary School & College — **GPA: 5.00 / 5.00**
Higher Secondary Certificate

Narayanganj Ideal School — **GPA: 5.00 / 5.00**
Secondary School Certificate

TECHNICAL SKILLS

Languages & Frontend: Python, TypeScript, SQL, C++, React, Vite, PWA

Backend & APIs: FastAPI, REST APIs, PostgreSQL, SQLAlchemy, SQLite, SQL Server, Google OAuth

Data & Tools: Pandas, Streamlit, Power BI, Power Query, Excel, Docker, Git/GitHub

Engineering: Object-Oriented Programming, Data Structures & Algorithms, Pytest, Vitest, Playwright

PROJECTS

Academic Assistant */ React, TypeScript, FastAPI, PostgreSQL, Google Classroom, PWA*

- **Purpose:** Give students one reliable place to review course tasks, deadlines, exams, notices, and completion status without modifying original Google Classroom content.
- Built a responsive PWA with browser-local dashboard snapshots, limited offline search, and queued state changes with conflict handling.
- Developed a FastAPI/PostgreSQL backend with Google OAuth, encrypted tokens, read-only Classroom synchronization, deterministic course matching, and optional Gemini explanations.
- Added server-side sessions, CSRF protection, user-scoped queries, migrations, and automated backend, frontend, and end-to-end tests.

Internship Application Tracker */ Python, SQL Server, pyodbc, OOP*

- **Purpose:** Centralize internship applications, companies, interviews, and progress so students can track their job-search pipeline consistently.
- Designed a normalized Student, Company, Application, and Interview schema with a reusable Python data-access layer.
- Implemented CRUD workflows, SQL LIKE keyword search, exception handling, and aggregated application-status reports.
- Used object-oriented modules to keep database operations and reporting maintainable and reusable.

Weather Dashboard */ Python, REST API, Pandas, Streamlit*

- **Purpose:** Turn live weather data into a simple decision-support view for current conditions, next-day forecasts, and previous searches.
- Integrated OpenWeatherMap data and calculated forecast minimum, maximum, and average temperatures with a trend chart.
- Used Pandas and CSV persistence to record searches and support filtering by city and date.
- Presented API responses through an interactive Streamlit interface with clear weather and forecast summaries.

Football Player Performance Dashboard */ SQL Server, Power Query, Power BI, Excel*

- **Purpose:** Transform raw football data into an interactive report for comparing performance, contributions, young talent, and future potential.
- Cleaned and transformed a Kaggle dataset with T-SQL, then modeled relationships and calculated fields in Power BI.
- Delivered four report pages with filters for nation, position, age, market value, goals, assists, and projections.
- Used Power Query and dashboard KPIs to make player comparisons and performance trends easier to explore.

LEADERSHIP, ACHIEVEMENTS & CERTIFICATIONS

Leadership	Deputy Coordinator, Department of Publication - AUST PIC; Member, Josephite Math Club
Achievement	GP FutureMakers 2026 - Advanced to Round 2 as one of 118 teams selected from 1,130 participating teams
Certifications	Google Prompting Essentials Specialization - Coursera; Data Analytics - Excel, Power BI, Python & SQL - Tutorials Point Bangladesh